2013
Interplanetary Small Satellite Conference
beyond LEO

Small Spinning Landers

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Pasadena, CA
2013 June 20-21

BLASTOFF!
2000
Artwork by Joe Bergeron
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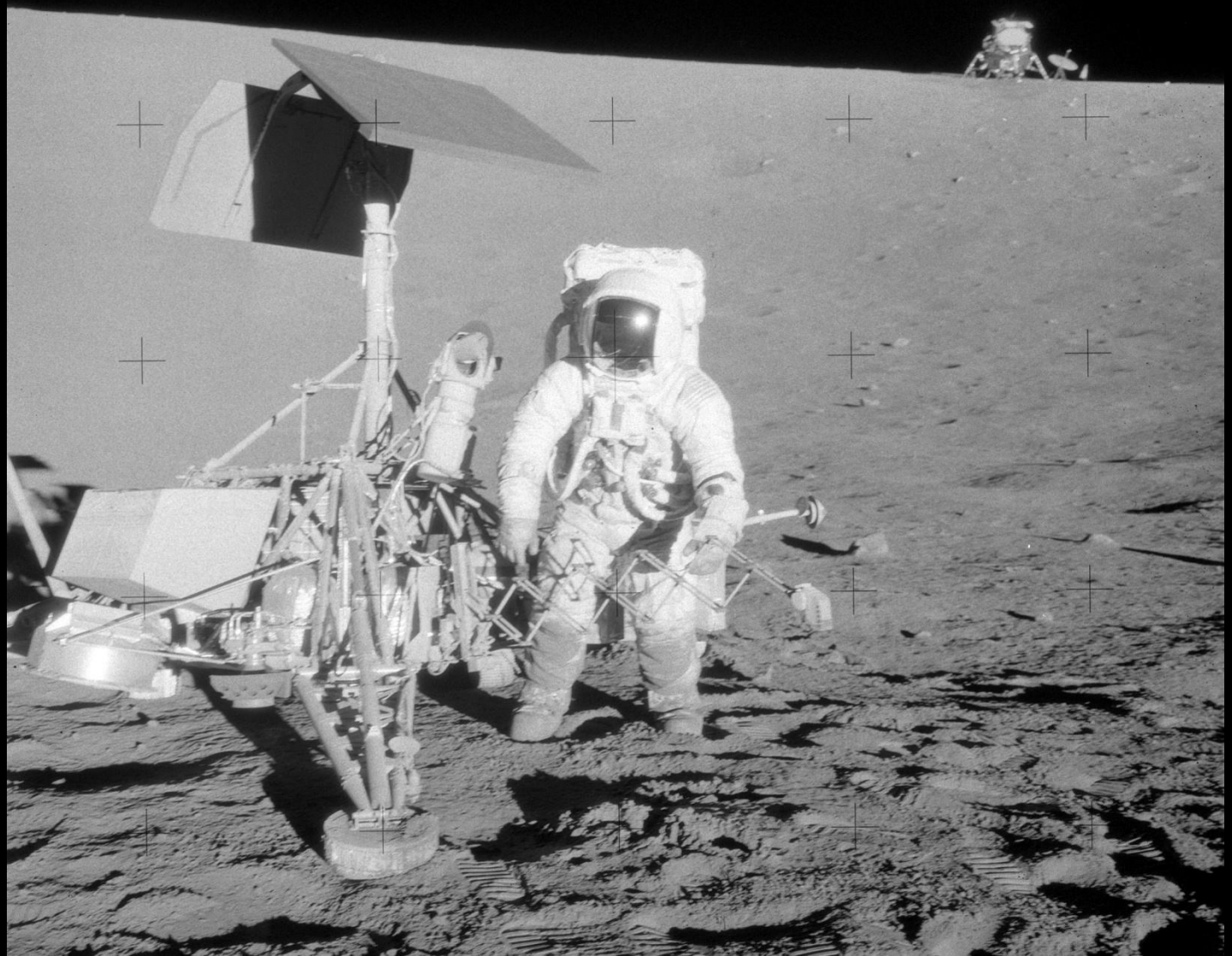
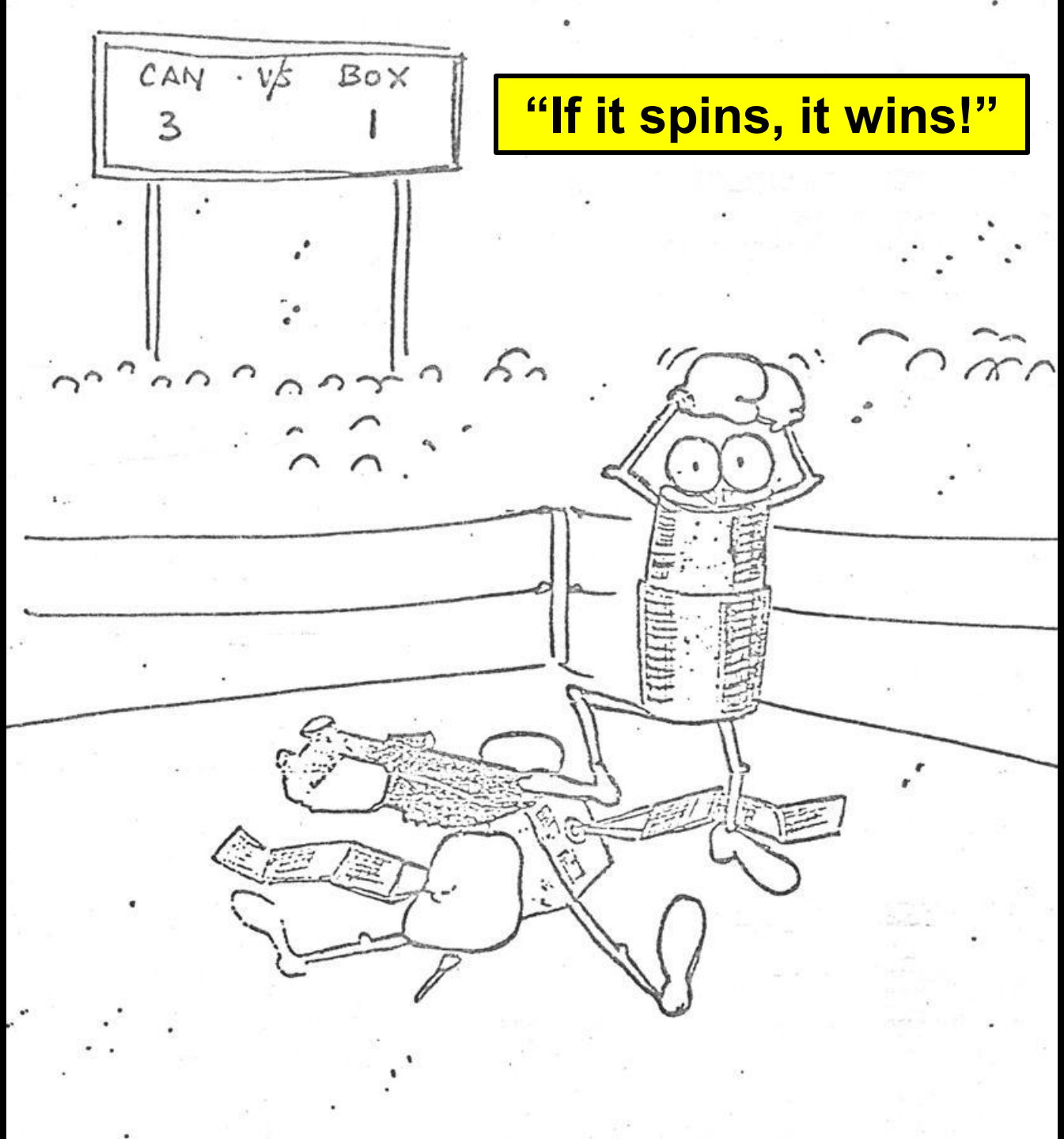
Late 2000

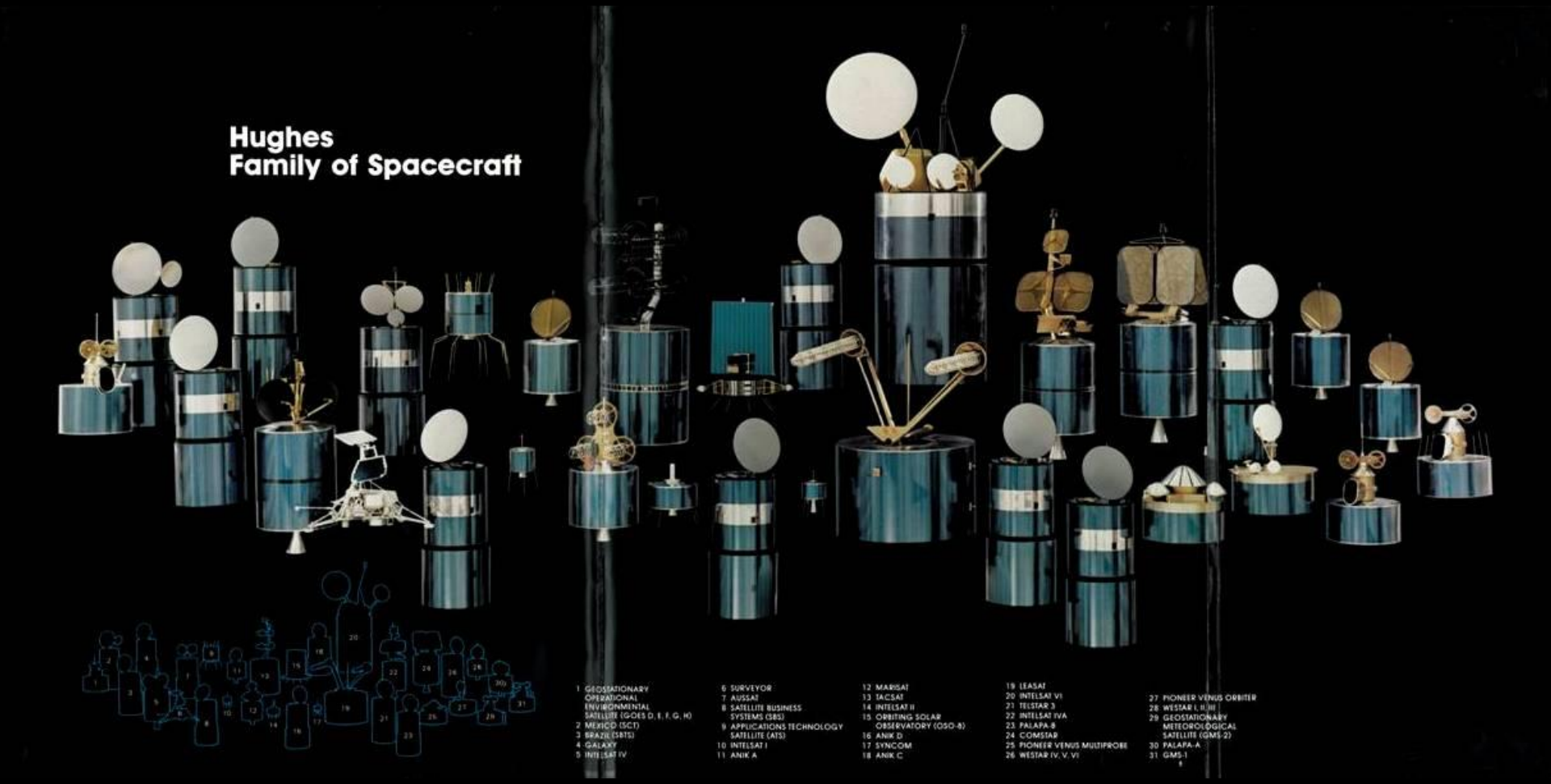
2007

1961

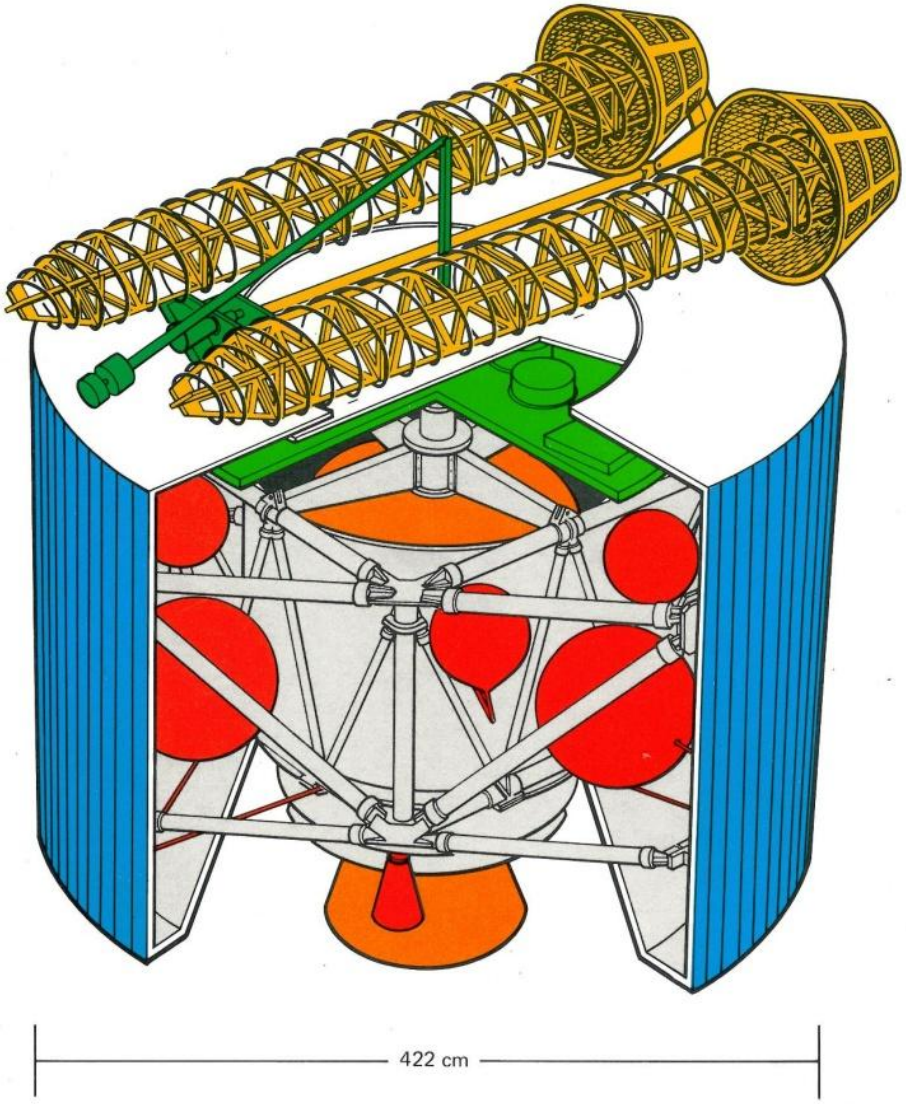




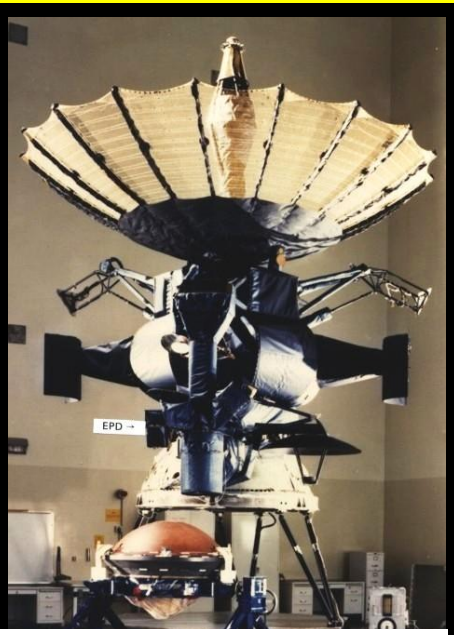
“If it spins, it wins!”

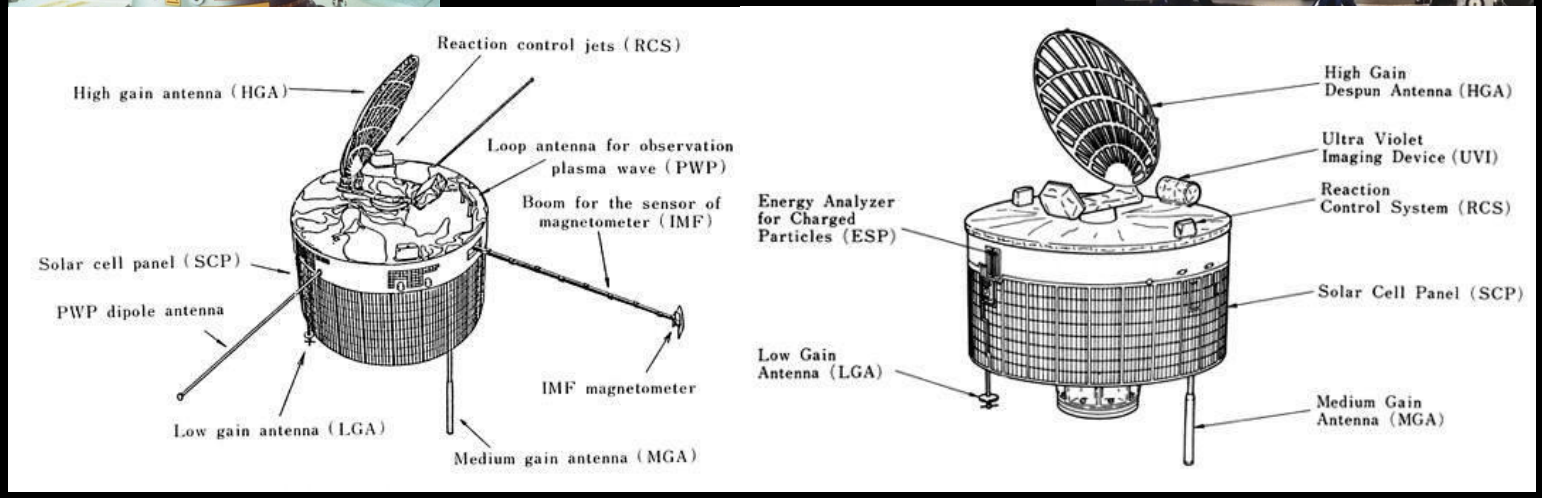

Hughes
Family of Spacecraft
1 GEOSTATIONARY OPERATIONAL ENVIRONMENTAL SATELLITE (G-OES D, E, F, G, H)
2 MEXICO (SCT)
3 BRAZIL (SBTS)
4 GALAXY
5 INTELSAT IV
6 SURVEYOR
7 AUSSAT
8 SATELLITE BUSINESS SYSTEMS (SBS)
9 APPLICATIONS TECHNOLOGY SATELLITE (ATS)
10 INTELSAT I
11 ANIK A
12 MARISAT
13 TACSAT
14 INTELSAT II
15 ORBITING SOLAR OBSERVATORY (OSO-8)
16 ANIK D
17 SYNCOM
18 ANIK C
19 LEASAT
20 INTELSAT VI
21 TELSTAR 3
22 INTELSAT IVA
23 PALAPA-B
24 COMSTAR
25 PIONEER VENUS MULTIPROBE
26 WESTAR IV, V, VI
27 PIONEER VENUS ORBITER
28 WESTAR I, II, III
29 GEOSTATIONARY METEOROLOGICAL SATELLITE (GMS-2)
30 PALAPA-A
31 GMS-1

1985


422 cm
UHF ANTENNAS
SHF ANTENNAS
SOLAR CELL ARRAY
DESPUN PLATFORM
LIQUID PROPELLANT SYSTEMS
SOLID PROPELLANT MOTOR

Dual-Spin Solar System Explorers





2007

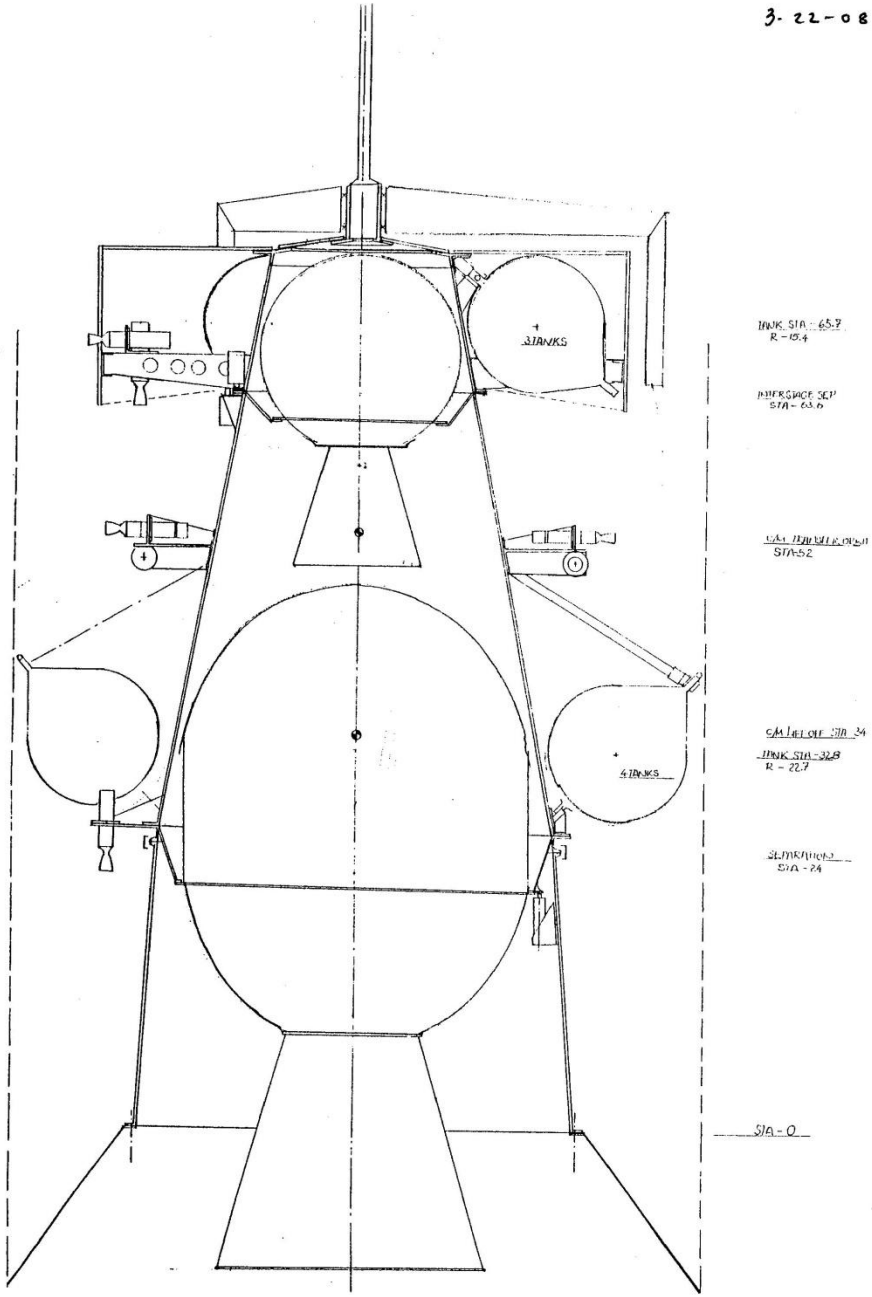
LUNAR XPRIZE
2008

OFFICIAL TEAM
SCSG
SOUTHERN CALIFORNIA SELENE GROUP
2008

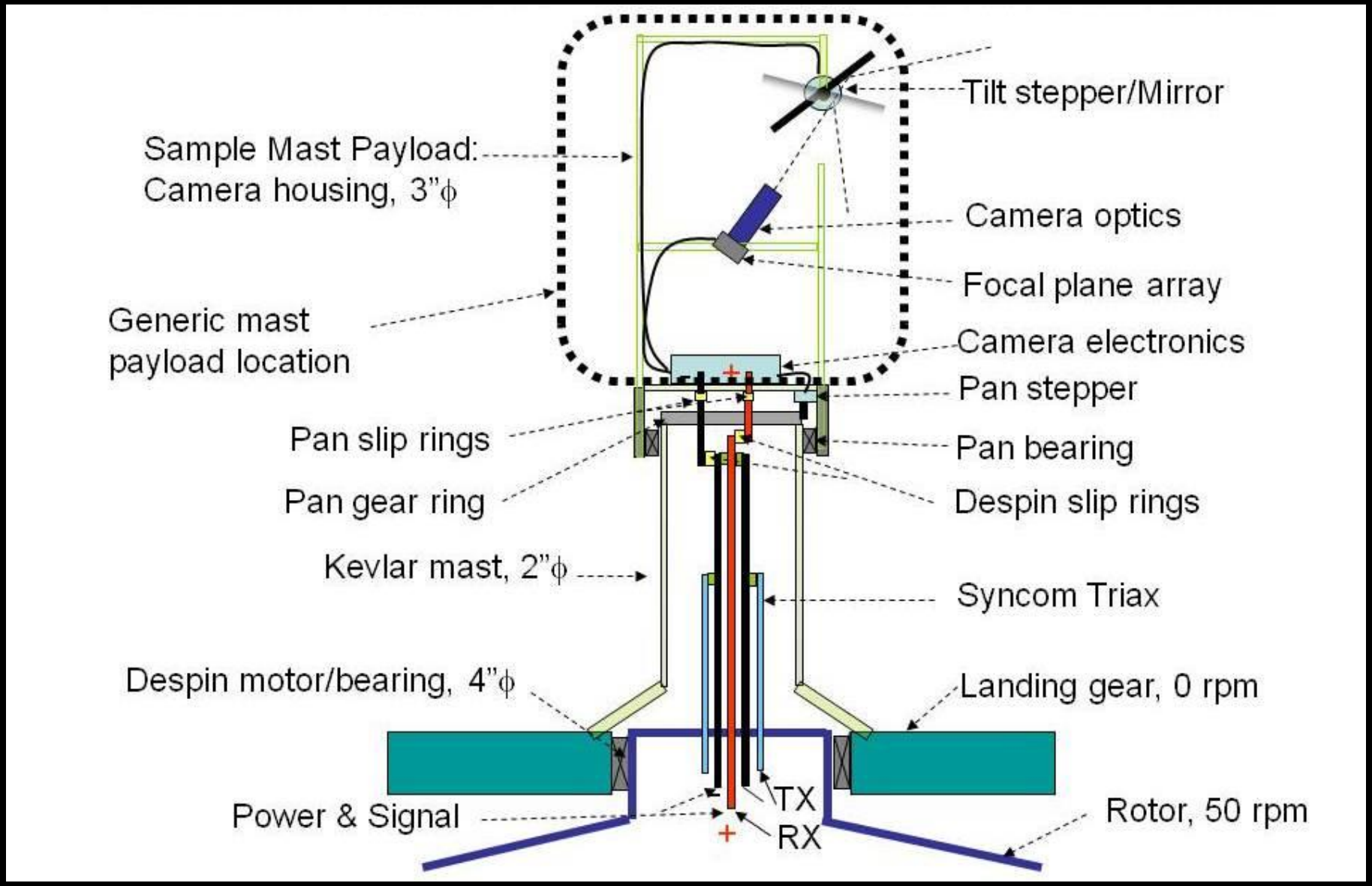
2008

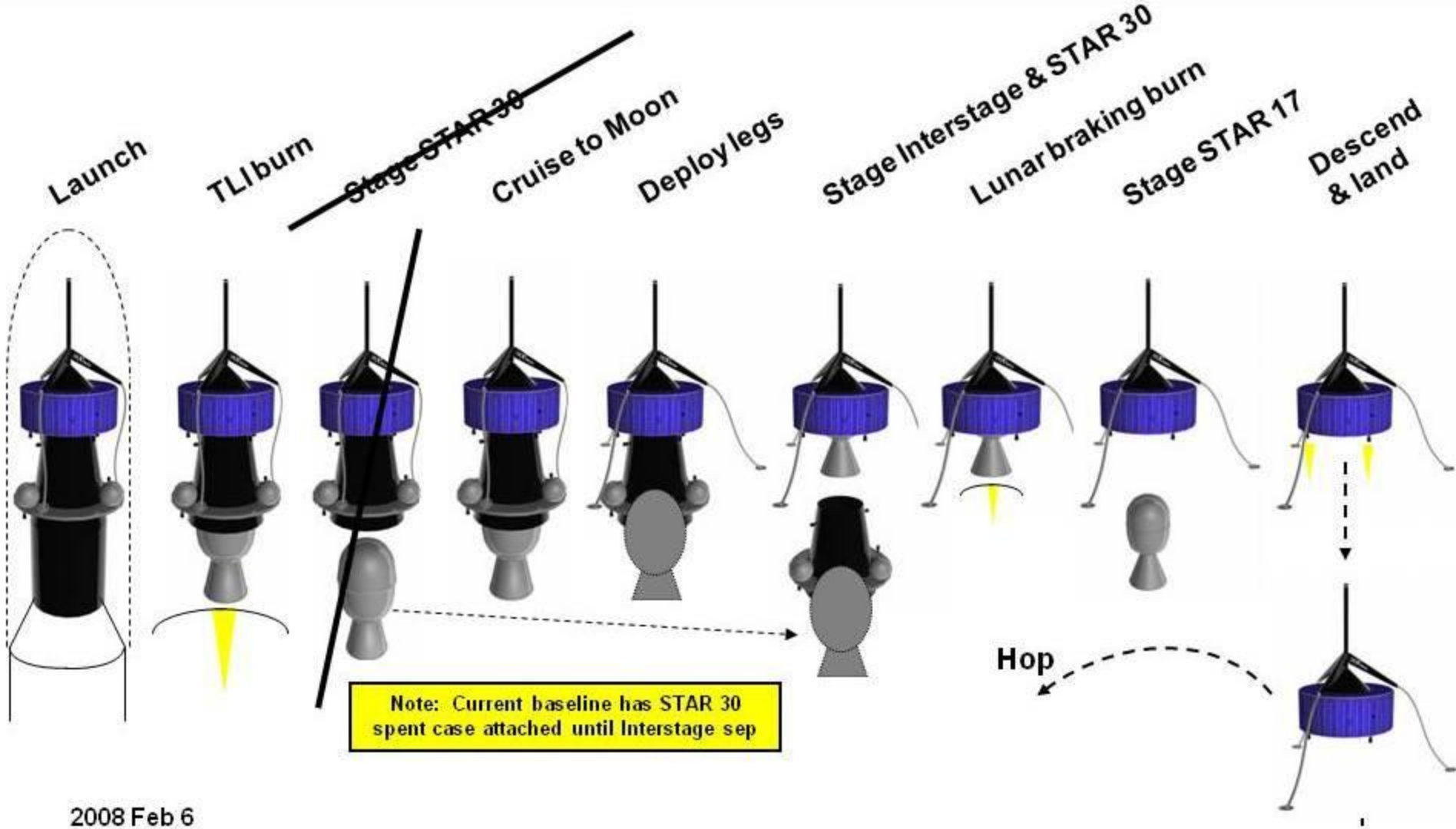
2008

Initial monoprop design



Initial monoprop design

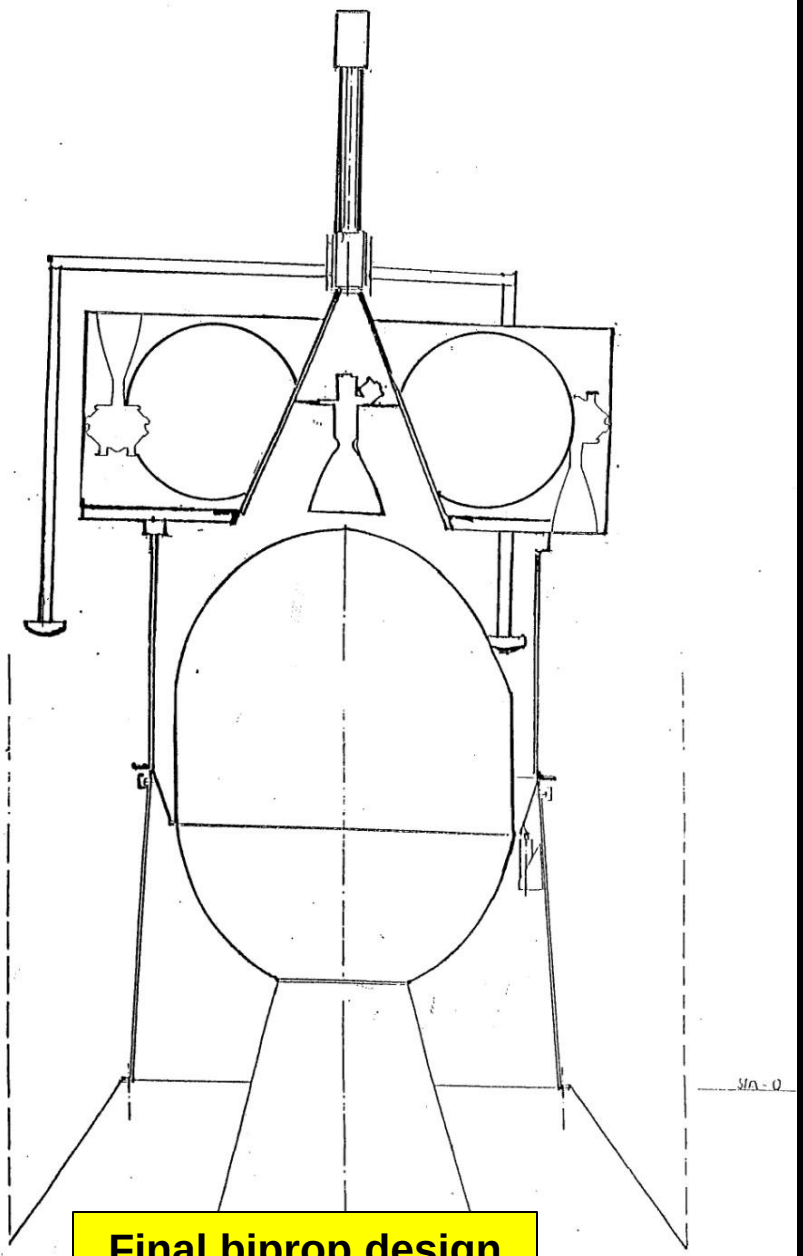

Tilt stepper/Mirror
Sample Mast Payload:
Camera housing, 3"ϕ
Camera optics
Focal plane array
Generic mast
payload location
Camera electronics
Pan stepper
Pan slip rings
Pan bearing
Pan gear ring
Despin slip rings
Kevlar mast, 2"ϕ
Syncom Triax
Despin motor/bearing, 4"ϕ
Landing gear, 0 rpm
TX
Power & Signal
RX
Rotor, 50 rpm


Launch
TLI burn
~~Stage STAR 30~~
Cruise to Moon
Deploy legs
Stage Interstage & STAR 30
Lunar braking burn
Stage STAR 17
Descend & land
Hop
Note: Current baseline has STAR 30 spent case attached until Interstage sep

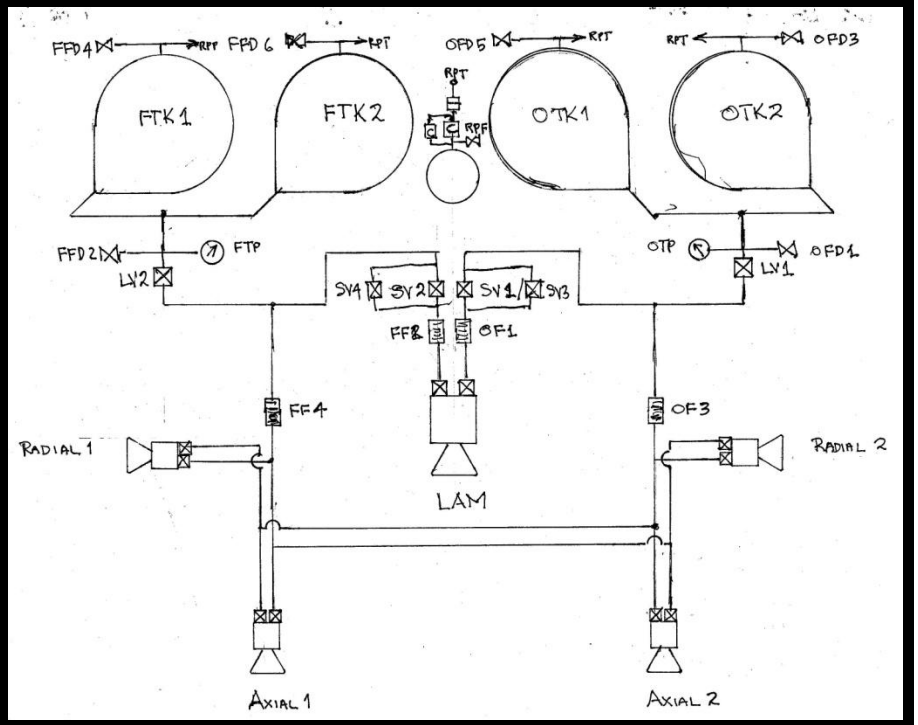
2008 Feb 6

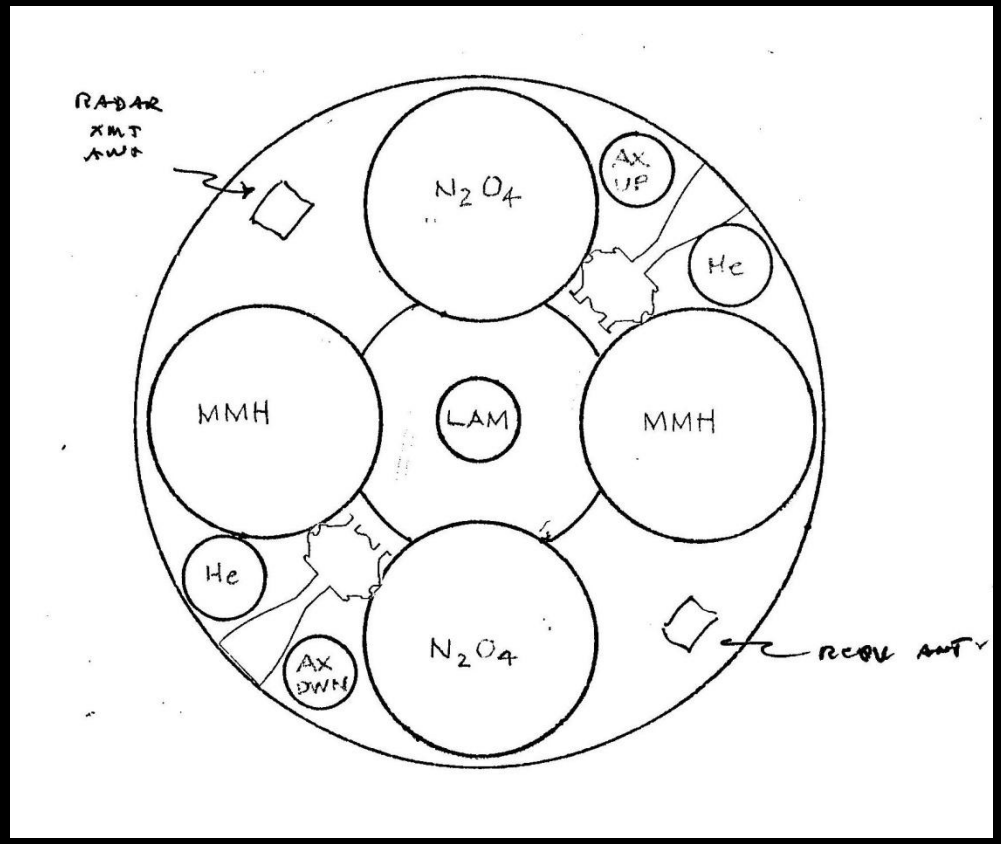
2.

passively stabilized.



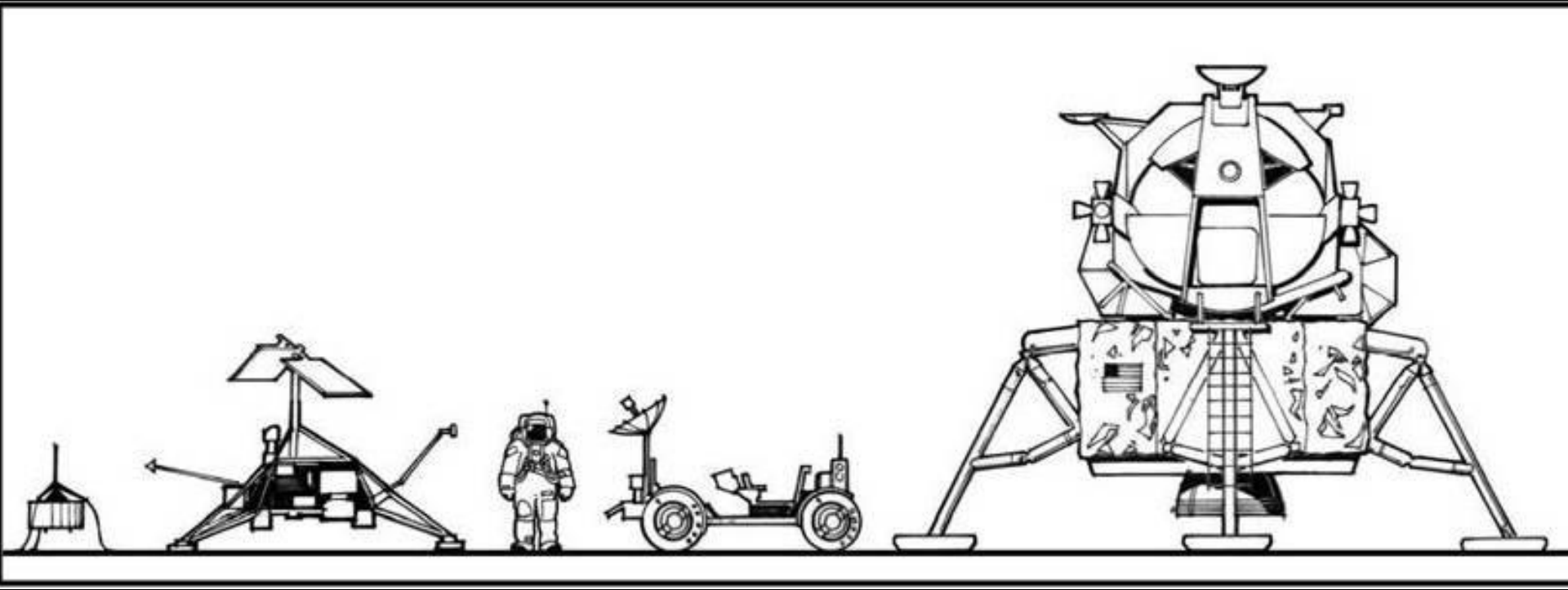
Final biprop design





Final biprop design

Size Comparison for GLXP Win



SCSG's GLXP Entry

- ~1/6 mass of Surveyor
- ~240 kg wet; ~50 kg dry
- >4.5 km/s ΔV capability

- Falcon 1e launch (900 kg to 200 km)
- Star 30 for TLI burn
- Biprop for all other maneuvers
- 90-hr cruise to Moon
- Land @ ~70° W, ~0° N
- Will not tip over
- ~5 km hopping capability

- $20-30 million total mission cost

Since Mid-2008

- **2008: SCSG team shuts down**
  - See *The Space Review*, 2013 June 3

  SpaceX discontinues Falcon 1 family
- **2009: Patents filed for spinning lander concept**
- **2010: Patents assigned to Ecliptic**
- **2011: Paper/presentation at LCPM-9 conf.**

  Multiple teaming discussions
- **2012: JPL-funded assessment contract**

  Paper/presentation at ASCE Extreme Env. conf.

  Paper/presentation at AIAA Space 2012

  U.S. patent secured

2013: Today's ISSC

Mission Applicability

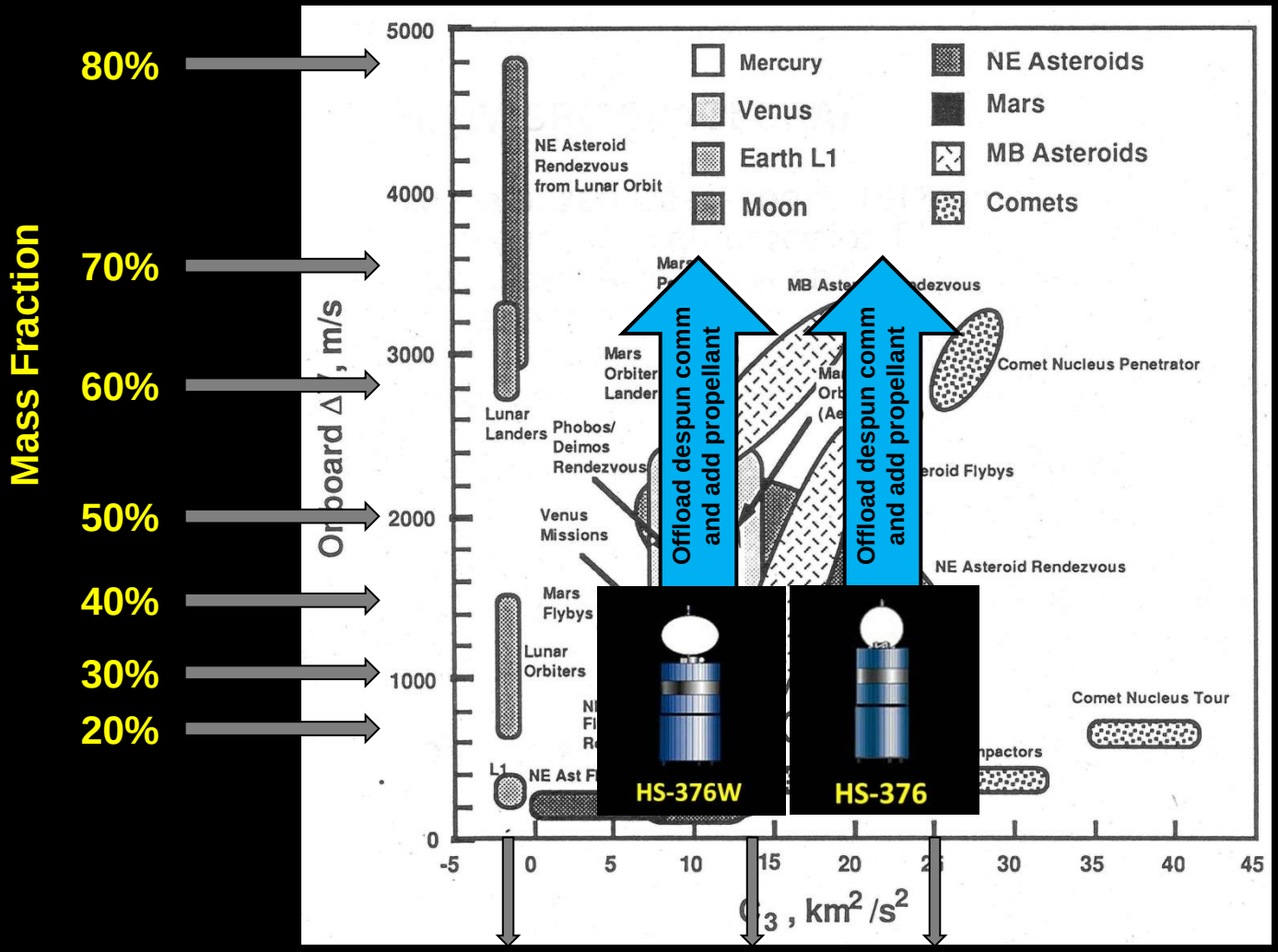
(E.g., Inner Solar System)

System Design	Sun	Mercury	Venus	Earth	Moon
Spinning Lander?	N/A				
Basic Landing System?	N/A				
Low-g Landing System?	N/A				
Aeroshell?	N/A				
Solar Arrays?	N/A				
RTGs/RHUs?	N/A		?		

	NEOs & Comets	Mars	Phobos & Deimos	Main Belt Asteroids
Spinning Lander?				
Basic Landing System?				
Low-g Landing System?				
Aeroshell?				
Solar Arrays?				
RTGs/RHUs?	?		?	?

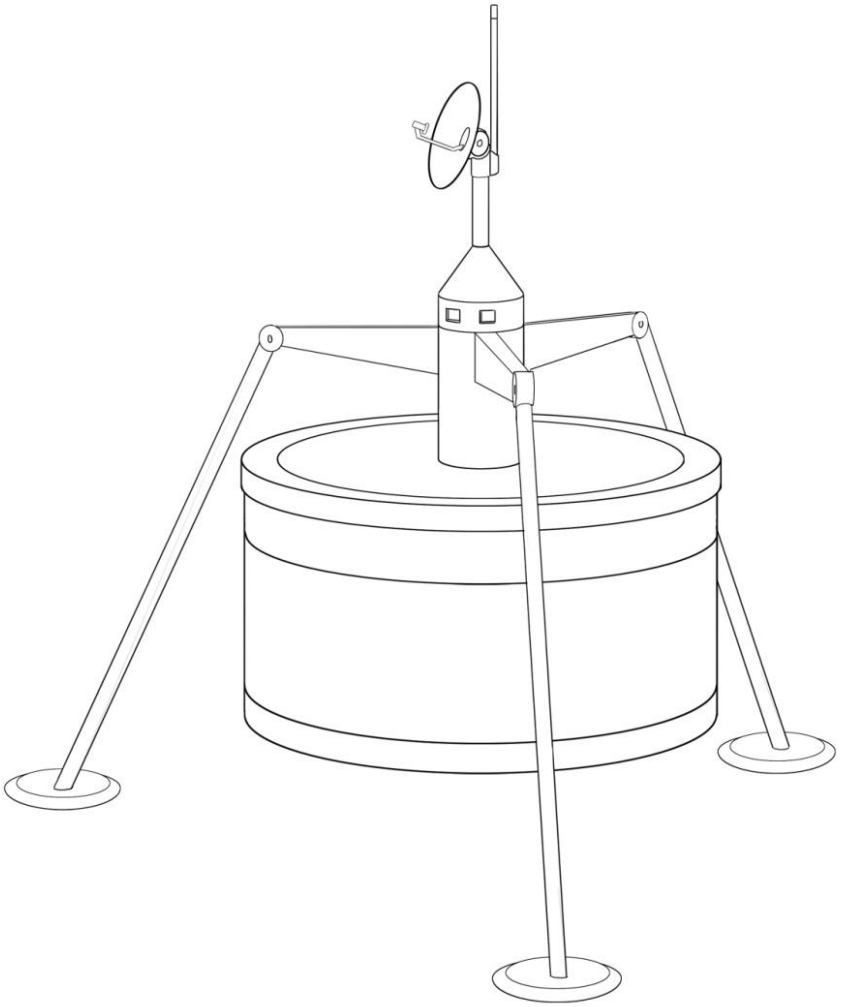
HS-376

Example: Falcon 9 Launch of HS-376

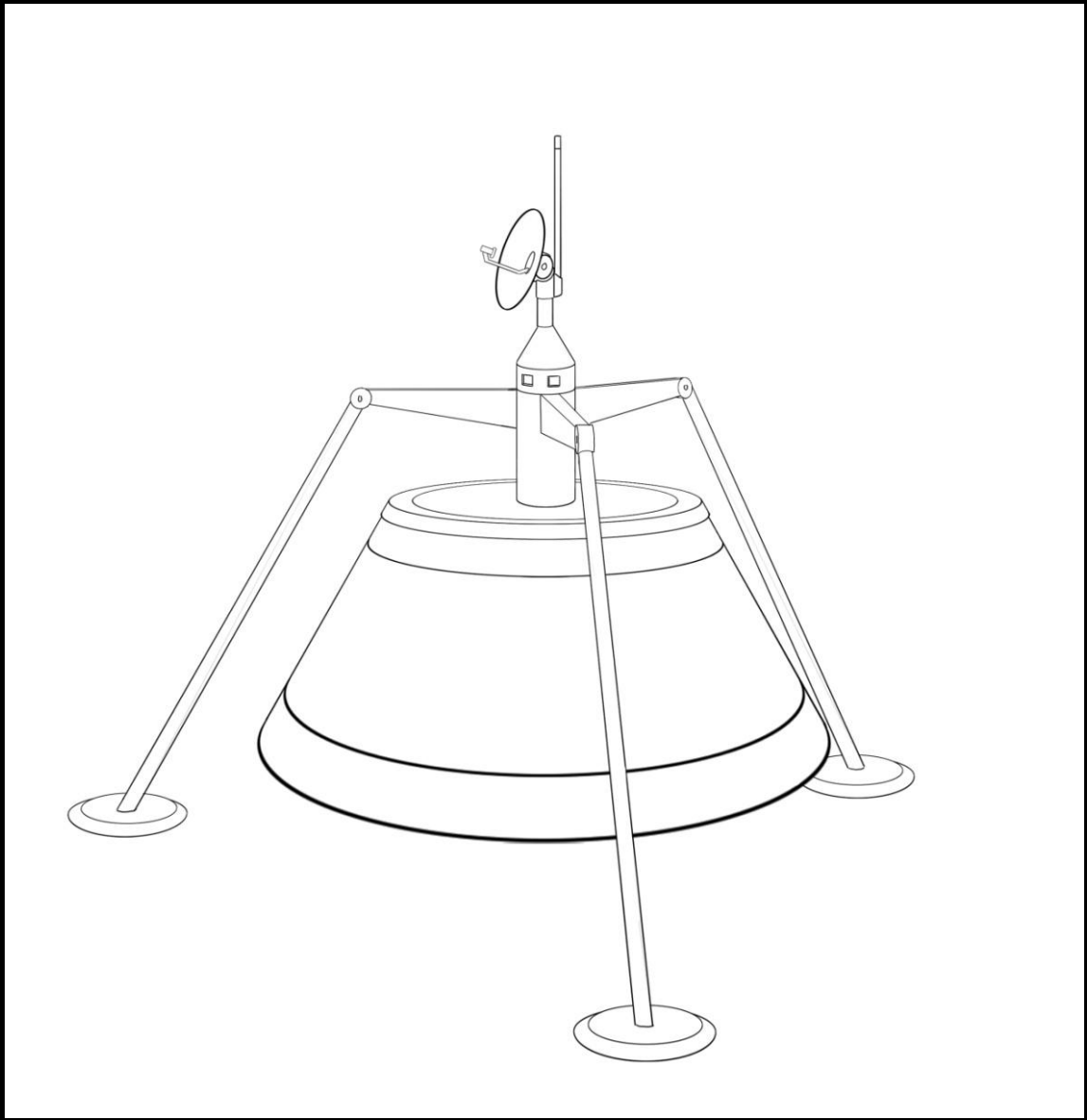


Falcon 9: ~2,500 kg ~1500 kg ~1,000 kg

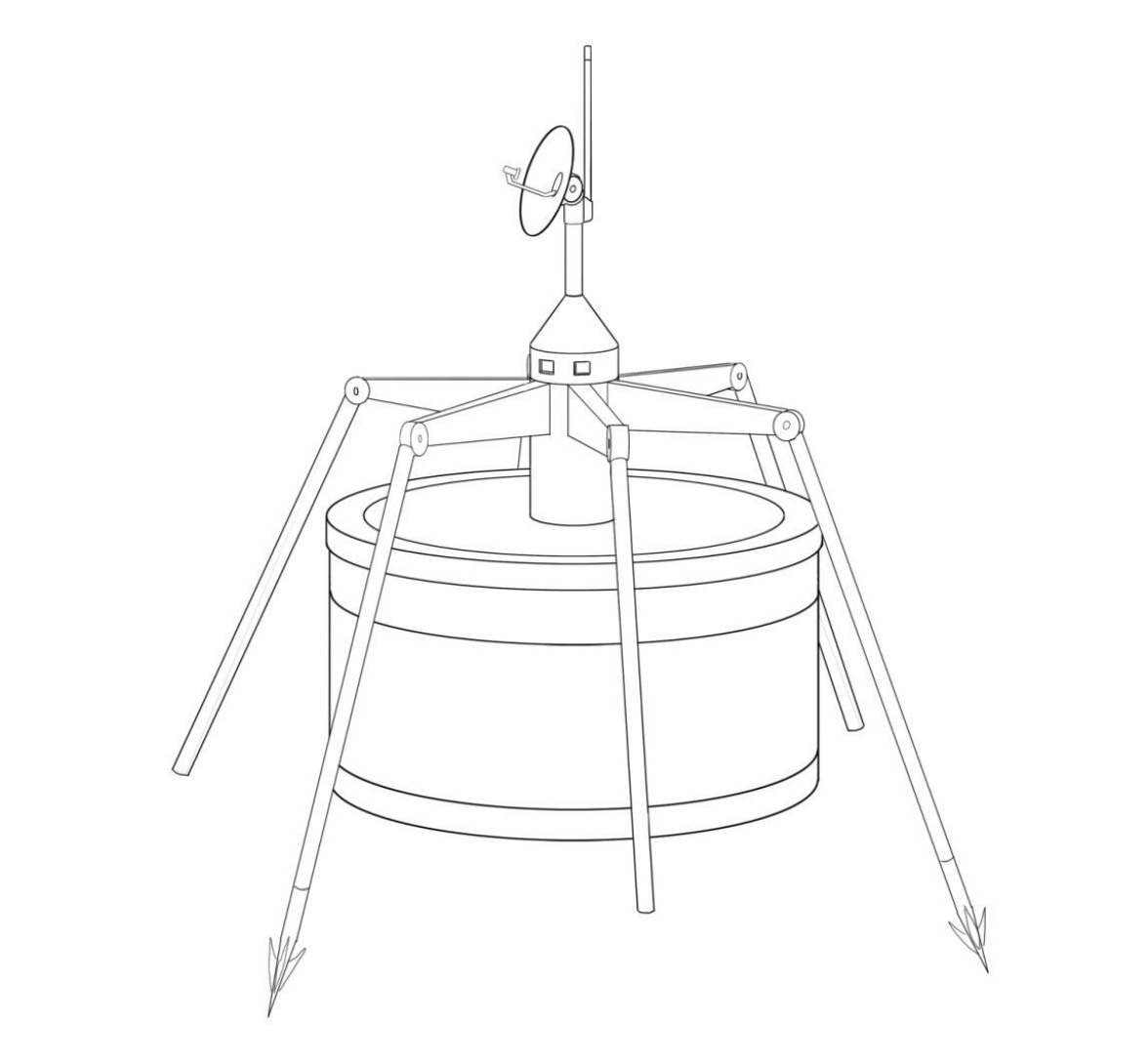
Spinning Lander for Moon



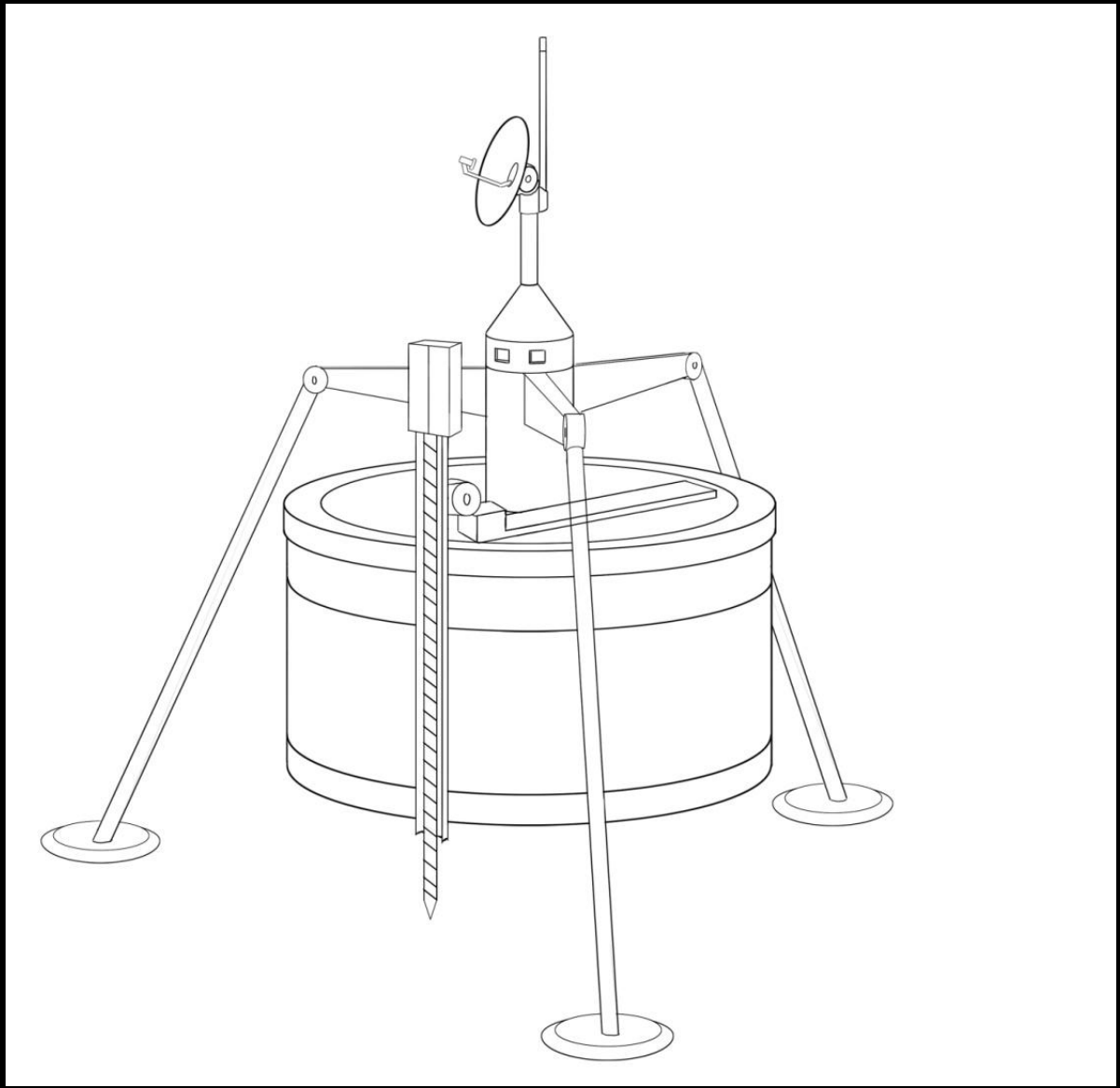
Spinning Lander for Mars



Spinning Lander for Small Bodies



Spinning Lander for Europa



Dual-Spin / 3-Axis Contrasts

	Dual-Spin	3-Axis
Primary mass balance	Critical (and MOIs, too)	Desired for some mission modes
Attitude stability	Inherent (passive)	Active
Attitude maneuverability	Slower and less efficent than 3-axis	Faster and more efficient than spinners
ACS sensors	Relatively simple (Earth, Sun, MIPs/IPs)	More complex
ACS software	4K words of PROM, 1 K of RAM, simple processor	64K words PROM, 64K of RAM, Mil-STD 1750A
ACS failures	Robust (power and temp safe); payload pointing may be lost	Can get into trouble quickly
Payload pointing performance	Coarse	Fine
Thruster control	1/2 as many as 3-axis	2x as many as spinners
Reaction mass	Spun structure (dual-spin requires BAPTA)	Reaction/momentum wheels, etc.
Propellant feed	Passive (centripetal force)	Prop management devices
Solar cell utilization for typical ops	~30%	~100%
Temp control	Mostly passive (BBQ mode); some heaters	Heaters, heat pipes, radiators
Overall flight software effort	~1/3 that of 3-axis	~3x that of spinners (esp. sims)

* Most inputs above from experienced Hughes (now Boeing) systems engineers who worked on dozens of dual-spin and 3-axis spacecraft

Key Attributes of Spinning Landers

- **Very strong heritage from dual-spin experience**
  - Esp. subsystem designs, scalability, ops
- **Robust attitude control**
- **Robust power system**
- **Benign thermal environment for most mission modes**
- **Simpler landing radar**
- **Robust landing system**
- **Robust hopping capability**
- **Potential for lower total system mass (vs. 3-axis)**
- **Potential for low end-to-end mission cost (vs. 3-axis)**

Some Areas for Further Research

- Subsystem scaling and sensitivity analyses
- Technology status and application vs. 10 to 30 years ago
- Extreme environment influences on design
- Attitude sensor tailoring
- Landing radar tailoring
- Landing operations and landing leg system tailoring
- Hopping operations
- Scientific and technology payloads
- Accommodation of robot arms
- Rover egress/ingress
- Integration into aeroshells
- Cargo accommodations
- Sample return systems
- Refueling and servicing provisions
- Alternative propulsion implementations
- Concept art

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